ModelIt!

Teacher's Guide

Why Do Animals Have Superpowers?

The Science of Body Structures and Survival

Quick Reference

Grade Level	Grade 4
NGSS Standard	4-LS1-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 4-LS1-1

Construct an argument that plants and animals have internal and external structures that function to support survival, growth, behavior, and reproduction.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how body structure type and environment danger level interact to determine survival rate.
Disciplinary Core Idea	**LS1.A Structure and Function** — Plants and animals have both internal and external structures that serve various functions in growth, survival, behavior, and reproduction.
Crosscutting Concept	**Structure and Function** — Students explore how the structure (shape, form, material) of an animal's body parts directly relates to their function (what job they do for survival).

CAST Testing Alignment

MULTIPLE CHOICE:

A scientist observes that desert foxes have very large ears while arctic foxes have very small ears. Which best explains this difference?

CONSTRUCTED RESPONSE:

Choose an animal and describe two body structures (one external and one internal) that help it survive. Explain how each structure's shape or design helps it do its job.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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> **For Your Confidence**
>
> This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Every animal on Earth is equipped with an amazing set of body structures — both inside and outside — that help it survive, find food, avoid danger, and raise its young. These structures are like built-in tools that have been perfected over millions of years of evolution.

External Structures

External structures are body parts on the outside that you can see. Examples include a bird's beak (shaped for the type of food it eats), a turtle's shell (armor against predators), a polar bear's thick fur (insulation against cold), a chameleon's color-changing skin (camouflage from predators), and a cheetah's long legs (speed for catching prey). Each structure's shape and design matches the specific job it needs to do.

Internal Structures

Internal structures are body parts inside the animal that you cannot see. A bird's hollow bones make it light enough to fly. A whale's enormous lungs let it hold its breath for over an hour while diving. A snake's flexible spine lets it squeeze through tight spaces. An eagle's sharp-focused eyes (with internal structures for magnification) let it spot a mouse from a mile away.

Structure Matches Environment

The key idea is that an animal's structures match the challenges of its environment. Desert animals have structures for conserving water and staying cool. Arctic animals have structures for staying warm and walking on ice. Forest animals have structures for climbing and hiding among trees. When the environment changes faster than animals can adapt, species can become endangered or extinct.

Biomimicry — Copying Nature's Designs

Engineers often study animal structures to inspire human inventions. Velcro was inspired by burrs that stick to animal fur. Bullet trains were redesigned after the kingfisher's beak to reduce noise. Shark skin patterns inspired swimsuits that reduce drag. This field is called biomimicry — 'bio' means life and 'mimicry' means copying. Animals have been solving engineering problems for millions of years!

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify body structure type and environment danger level as external components and survival rate as an internal component that responds to how well structures match dangers.

E **Establish Relationships**
Students determine that better structure-environment matches positively affect survival rate, and that mismatches reduce survival rate.

V **Visualize & Model**
Students build a model showing how different structure types perform differently depending on the environment's danger profile.

E **Evaluate Outcomes**
Students run perfect match, mismatch, and high danger scenarios to observe how the relationship between structure and environment determines survival.

R **Revise & Extend**
Students add food availability or predator speed to explore more complex survival challenges.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"If you could have any animal's superpower, what would it be? A cheetah's speed? An eagle's eyesight? A chameleon's camouflage?"*

Watch For

Let students share their animal superpower picks. Then reveal: those are not superpowers — they are STRUCTURES.

Slide 2: Learning Objectives

3 min

Say: *"Today we are going to discover why animals have the amazing body features they do."*

Watch For

Have students read objectives aloud. Pre-teach 'structure' and 'function' using a pencil as an example (structure = shape, function = writing).

Slide 3: Big Question

3 min

Say: *"Animals do not have backpacks, houses, or jackets. They survive using ONLY their body. How is that possible in dangerous environments?"*

Watch For

Show images of extreme environments and ask: What would an animal NEED to survive here?

Slide 4: LEVER Framework

2 min

Say: *"We are going to build a model to figure out why certain animals survive where others would not."*

Watch For

Preview LEVER steps. Remind students that models help us test ideas without going to dangerous places.

Slide 5: Activity 1: Components

8 min

Say: *"What determines whether an animal survives? Think about the animal AND its environment."*

Watch For

Guide sorting. Explain that both body structure type and environment danger level are external — they are set before the model runs.

Slide 6: Activity 2: Connections

8 min

Say: *"When an animal's body matches its environment's dangers, survival goes UP. But what if there is a mismatch?"*

Watch For

Help students draw arrows. Discuss how the same structure can help or hurt depending on the environment.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let us test it! What happens when we put a thick-furred animal in the desert?"*

Watch For

Students predict first, then run scenarios. The mismatch scenario is the most surprising result.

Slide 8: Discoveries

5 min

Say: *"Every animal superpower is actually a survival structure that MATCHES its environment!"*

Watch For

Show real examples: polar bear fur, cactus spines, eagle talons. Ask students to identify the structure and its function.

Slide 9: STEM Challenge

12 min

Say: *"Now YOU are nature's engineers! Design the perfect animal for an extreme environment."*

Watch For

Assign or let groups choose extreme environments. Groups design and draw animals with labeled survival structures.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Body Structure Type, Environment Danger Level
- **INTERNAL:** Survival Rate

Relationships

- **Body Structure Type to Survival Rate:** POSITIVE (+) when matched — When an animal's body structures match the dangers in its environment (like thick fur in the cold), survival rate is high. The better the match between structure and environment, the more likely the animal is to survive and thrive.
- **Environment Danger Level to Survival Rate:** NEGATIVE (-) — More dangers in the environment make it harder for all animals to survive. Higher danger means lower survival rate unless the animal has structures that specifically counter those dangers.

Key Discoveries

- Animals with body structures that match their environment's dangers have much higher survival rates
- The same body structure can be helpful in one environment but harmful in another
- Both external structures (claws, shells, fur) and internal structures (strong bones, efficient lungs) are critical for survival
- Animals in more dangerous environments tend to have more specialized survival structures

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Animals choose to grow the structures they need"	Animals cannot choose their body structures — they are born with them based on their genes from their parents. Over millions of years, animals with better-matched structures survived more often and had more babies, passing those useful features to the next generation. This is called natural selection.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"All animals in the same environment look the same"	Animals in the same environment can have VERY different structures that solve the same problem in different ways. In the Arctic, polar bears have thick fur for warmth while walruses have thick blubber. Both stay warm, but the structure is totally different.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Body structures are only on the outside"	Many of the most important survival structures are INSIDE the animal where you cannot see them. A bird's hollow bones let it fly. A whale's giant lungs let it dive for hours. A snake's flexible spine lets it squeeze through tight spaces. Both internal AND external structures are essential for survival.